# README: Extension calculations

Normal reduction, $\mathrm{Ext}^1_\tau$, $\mathrm{Ext}_{0,\tau}$ and the exact sequence

The calculator computes the $t$-actions and maps in the exact sequence of Theorem 6.1 (see Section 4):

$$0 \longrightarrow \mathrm{Ext}_{0,\tau}(\Phi,\Psi) \xrightarrow{\iota} \mathrm{Ext}^1_\tau(\Phi,\Psi) \xrightarrow{p} (Q, \theta\ \mathrm{id}_Q + \bar{N}_\Psi) \longrightarrow 0.$$

## 1. Quick start

**Symbolic mode.** Select `None` to leave the positive characteristic and prime power $q$ unspecified. Over $\mathbb{F}_q(\theta)$, Frobenius fixes $\mathbb{F}_q$ and sends $\theta$ to $\theta^q$; its iterates and the Ore rule $\tau a = a^q \tau$ are retained symbolically.

**Concrete mode.** For arithmetic over $\mathbb{F}_q(\theta)$ with $q = p^{\mathtt{n}}$:

1. For a concrete computation, leave the `None` checkbox unselected. Set `prime` to a prime $p$ and `n` to a positive integer. The Frobenius is $q = p^{\mathtt{n}}$; the coefficient characteristic is $p$.
2. Set the matrix sizes: $\Phi_t$ is $d \times d$ and $\Psi_t$ is $e \times e$. These sizes are independent of the exponent `n` above.
3. Enter the two matrices in the mathematical editors or the corresponding `LaTeX` boxes. For the scalar example on page 5, take $p = 3$, $\mathtt{n} = 1$, $d = e = 1$ and enter:

```
Phi_t:  \begin{pmatrix}\theta+\tau^2\end{pmatrix}
Psi_t:  \begin{pmatrix}\theta+\tau\end{pmatrix}
```

4. Click `Run` and wait for `COMPLETE`. After an input change, `RESULTS STALE` means the displayed matrices belong to an earlier calculation. Run again before using or copying them.

## 2. The output panels at a glance

Write $M = \dim \mathrm{Ext}^1_\tau(\Phi,\Psi)$, $h = \dim_L Q$ and $M_0 = M - h = \dim \mathrm{Ext}_{0,\tau}(\Phi,\Psi)$, with $Q$ as in Section 4. These are dimensions of the underlying additive groups. The labels `Ext1` and `Ext0` in the interface mean $\mathrm{Ext}^1_\tau$ and $\mathrm{Ext}_{0,\tau}$, respectively.

| Output label | Size | Meaning |
|---|---|---|
| Output 1: Ext1 Matrix | $M \times M$ | $\Pi_t$: the $t$-action on full normal coordinates of $\mathrm{Ext}^1_\tau(\Phi,\Psi)$. |
| Output 2: Ext0 Matrix | $M_0 \times M_0$ | $\Pi_{0,t}$: the induced $t$-action on $\mathrm{Ext}_{0,\tau}(\Phi,\Psi)$. |
| Output 3: K Matrix | $M \times M_0$ | $K$: inclusion; $y$ in $\mathrm{Ext}_{0,\tau}$ coordinates becomes $Ky$ in full coordinates. |
| Output 4: R Matrix | $M_0 \times M$ | $R$: auxiliary matrix for extracting $\mathrm{Ext}_{0,\tau}$ coordinates; $RK = I_{M_0}$. |
| Output 5: P Matrix | $h \times M$ | $P$: quotient projection; a full coordinate vector $x$ lies in $\mathrm{Ext}_{0,\tau}$ exactly when $Px = 0$. |
| Output 6: S Matrix | $e \times d$ | $S = \varepsilon$: auxiliary matrix recording zero/nonzero constant inner directions. |

Use `Copy As LaTeX` next to an output to copy its matrix. The `Compatible Basis` panel displays the jump matrix (labelled `U`) and its inverse; we denote them by $C$ and $C^{-1}$, as in the paper. The `Leaf Count` and `Byte Count` metrics measure expression size and storage, not dimensions or ranks.

# 3. Algebra, assumptions and normal coordinates

## The coefficient convention

Let $L$ be a coefficient field of characteristic $p$ containing $\mathbb{F}_q$, with characteristic parameter $\theta \in L$. The symbol $t$ is the generator of $\mathbb{F}_q[t]$; $\theta$ is its image under the characteristic map. The operator $\tau$ acts by $c \mapsto c^q$.

Calculations take place in the Ore ring $L\{\tau\}$, with coefficients on the left:

$$\tau a = a^q \tau, \qquad (a\tau^i)(b\tau^j) = a\, b^{q^i} \tau^{i+j}.$$

Thus $\sum_j a_j \tau^j$ represents the additive polynomial $c \mapsto \sum_j a_j c^{q^j}$. Products of output action matrices mean composition of additive maps, using this Ore rule.

## Admissible input for this reduction

Write

$$\Phi_t = \theta I_d + N_\Phi + \sum_{r \geq 1} A_r \tau^r, \qquad \Psi_t = \theta I_e + N_\Psi + \sum_{r \geq 1} B_r \tau^r.$$

The matrices $N_\Phi$ and $N_\Psi$ must be nilpotent. The positive coefficient blocks of $\Phi_t$ must admit the compatible jump basis required by the reduction algorithm. It selects columns

$$u_s = A_{j_s} \mathbf{e}_s, \qquad C = (u_1 \ \cdots \ u_d), \qquad j_s \geq 1,$$

with $C$ invertible and a compatible column ordering $\sigma$. The selected columns must respect the successive rank jumps of the positive coefficient blocks; an arbitrary invertible choice is insufficient. The degree hypothesis is

$$\deg_\tau \Psi_t < \min_{1 \leq s \leq d} j_s.$$

For a scalar $\Phi_t = \theta + \tau^a$, this reads $\deg_\tau \Psi_t < a$.

## What is reduced

A biderivation is determined by its value $V = \delta(t) \in \mathrm{Mat}_{e \times d}(L\{\tau\})$. Its inner changes have the form

$$\delta_H(t) = H\Phi_t - \Psi_t H, \qquad H \in \mathrm{Mat}_{e \times d}(L\{\tau\}).$$

Accordingly, extension classes are represented by these matrices modulo inner changes. Under the reduction hypotheses, full normal representatives satisfy

$$\deg_\tau V_{is} < j_s.$$

The full normal coordinate family is

$$\mathcal{E} = \{E_{is}\tau^r : 1 \leq i \leq e,\ 1 \leq s \leq d,\ 0 \leq r < j_s\}, \qquad J = \sum_{s=1}^{d} j_s, \qquad M = eJ.$$

The triples $(i, s, r)$ are ordered lexicographically, with the degree varying fastest. Here $E_{is}$ is an $e \times d$ elementary matrix.

## How Output 1 is computed

The $t$-action on a class is $t \cdot [V] = [\Psi_t V]$. For each normal coordinate direction, the algorithm reduces $\Psi_t(cE_{is}\tau^r)$ with a formal coefficient $c$. If an output coordinate is $\sum_j a_j c^{q^j}$, the corresponding entry of $\Pi_t$ is $\sum_j a_j \tau^j$. Keeping $c$ formal is essential: substituting $c = 1$ would lose the Frobenius exponents.

# 4. $\mathrm{Ext}_{0,\tau}$ and the exact sequence

## The constant-term obstruction

As in Section 6 of the paper, put

$$W = \mathrm{Mat}_{e\times d}(L), \qquad T : W \longrightarrow W, \qquad T(H_0) = H_0 N_\Phi - N_\Psi H_0.$$

For an inner change, $(H\Phi_t - \Psi_t H)_0 = T(H_0)$. Thus equivalent representatives have constant coefficients differing by an element of $\operatorname{im} T$. Define

$$Q = W/\operatorname{im} T, \qquad r = \operatorname{rank}_L T, \qquad h = \dim_L Q = ed - r.$$

Left multiplication by $N_\Psi$ preserves $\operatorname{im} T$, since $N_\Psi T(H_0) = T(N_\Psi H_0)$. It induces the nilpotent map

$$\bar{N}_\Psi : Q \longrightarrow Q, \qquad \bar{N}_\Psi(C_0 + \operatorname{im} T) = N_\Psi C_0 + \operatorname{im} T.$$

## Theorem 6.1: the exact sequence of $t$-modules

Exactly as in Theorem 6.1 of *Computing extensions of flag-jump $t$-modules*, the sequence is

$$0 \longrightarrow \mathrm{Ext}_{0,\tau}(\Phi,\Psi) \xrightarrow{\iota} \mathrm{Ext}^1_\tau(\Phi,\Psi) \xrightarrow{p} (Q, \theta\,\mathrm{id}_Q + \bar{N}_\Psi) \longrightarrow 0.$$

The quotient action uses the induced endomorphism $\bar{N}_\Psi$ defined above. In the program, $K$ is the coordinate matrix of $\iota$, and $P$ is the coordinate matrix of $p([V]) = V_0 + \operatorname{im} T$. The subgroup $\mathrm{Ext}_{0,\tau}(\Phi,\Psi)$ consists of classes admitting a representative with zero constant coefficient. For a full normal representative,

$$[V] \in \mathrm{Ext}_{0,\tau}(\Phi,\Psi) \quad\Longleftrightarrow\quad V_0 \in \operatorname{im} T.$$

Its normal constant coefficient need not itself vanish: subtracting a constant inner change removes it but can introduce higher powers of $\tau$.

## Dimensions and normal coordinates

The dimensions of the three terms are $M - h$, $M$ and $h$. In particular,

$$\boxed{M_0 = e\sum_s (j_s - 1) + r = M - h.}$$

A coordinate basis of $\mathrm{Ext}_{0,\tau}$ consists of a basis of $\operatorname{im} T$ in degree zero, followed by all positive normal monomials $E_{is}\tau^a$ with $1 \le a < j_s$, in their order from $\mathcal{E}$. Thus $K$ may mix constant coordinates. Although $h = \dim_L \ker T$, the quotient is $Q = \operatorname{coker} T$.

## The exact sequence in the program's coordinates

In a chosen basis of $Q$, let $Q_t$ be the matrix of $\theta\,\mathrm{id}_Q + \bar{N}_\Psi$. The program's sequence is

$$0 \longrightarrow (\mathbb{G}_a^{M_0}, \Pi_{0,t}) \xrightarrow{K} (\mathbb{G}_a^{M}, \Pi_t) \xrightarrow{P} (\mathbb{G}_a^{h}, Q_t) \longrightarrow 0.$$

The program's abbreviated action labels $\Pi_0$, $\Pi$ and $Q$ mean $\Pi_{0,t}$, $\Pi_t$ and $Q_t$, respectively. The quotient space $Q$ above is distinguished from its action matrix $Q_t$. This matrix is constant in $\tau$ and is $\theta I_h$ plus a nilpotent matrix; it equals $\theta I_h$ when $N_\Phi = N_\Psi = 0$. The six output panels do not include a separate $Q_t$ panel; $S$ is a constant-direction diagnostic.

# 5. Coordinate maps and auxiliary matrices

## How K, R and P fit together

For constant matrices, vec stacks columns. These coordinates are inserted into the constant positions of the lexicographic full basis $\mathcal{E}$. The matrix of $T$ is

$$D = N_\Phi^T \otimes I_e - I_d \otimes N_\Psi, \qquad \operatorname{vec}(T(H_0)) = D \operatorname{vec}(H_0).$$

Choose $B$ with columns forming a basis of $\operatorname{im} D$, and complete it by $Z$ to an invertible matrix $(B\ Z)$. In the paper's notation, put

$$\begin{pmatrix} U \\ P_0 \end{pmatrix} = (B\ Z)^{-1}, \qquad c = \operatorname{vec}(V_0).$$

If $x_+$ lists the positive normal coordinates, the coordinate rules are

$$\begin{aligned} K(y_{\mathrm{im}}, y_+): \quad (c, x_+) &= (By_{\mathrm{im}}, y_+), \\ Rx = (Uc, x_+), &\qquad\qquad Px = P_0 c. \end{aligned}$$

Let $Z_{\mathrm{full}}$ insert $Z$ into the full constant positions. Then

$$PK = 0, \quad RK = I_{M_0}, \quad PZ_{\mathrm{full}} = I_h, \quad KR + Z_{\mathrm{full}}P = I_M.$$

For $x \in \ker P$, $Rx$ recovers the unique $\mathrm{Ext}_{0,\tau}$ coordinates. The maps $R$ and $Z_{\mathrm{full}}$ need not be $t$-module morphisms, so these identities do not assert a splitting of the exact sequence.

## Actions and Frobenius twists

All products involving $\tau$ use Ore multiplication. With $\Pi_t = \sum_a \Pi_a \tau^a$, the maps and actions satisfy

$$\Pi_t K = K\Pi_{0,t}, \qquad P\Pi_t = Q_t P,$$

$$\boxed{\Pi_{0,t} = R\Pi_t K = \sum_a R\Pi_a K^{(q^a)} \tau^a.}$$

$$Q_t = \theta I_h + P_0(I_d \otimes N_\Psi)Z = P\Pi_t Z_{\mathrm{full}}.$$

The superscript $(q^a)$ means entrywise Frobenius. These twists are essential when $K$ has coefficients outside the Frobenius-fixed field. The formula using $N_\Psi$ computes the quotient action directly from constant matrices.

## Output 6: the auxiliary matrix S

For the jump matrix $C$, put $U_{is} = E_{is}C^{-1}$, as in the paper. Then

$$S_{is} = \varepsilon_{is} = \begin{cases} 0, & U_{is}N_\Phi - N_\Psi U_{is} = 0, \\ 1, & U_{is}N_\Phi - N_\Psi U_{is} \neq 0. \end{cases}$$

Here $S$ is the program's diagnostic matrix $\varepsilon$; the paper also uses $S$ for a coordinate-permutation matrix. Its zero entries identify independent directions in $\ker T$, but other linear combinations may also lie in this kernel. Thus counting zeros need not give $h$. Compute $h = ed - \operatorname{rank} T$ and $M_0 = M - h$ using the full map $T$; see the $J_2$ example on page 6.

## 6. A complete scalar example

Choose $p = 3$, $\mathtt{n} = 1$, hence $q = 3$, and take

$$\Phi_t = \theta + \tau^2, \qquad \Psi_t = \theta + \tau.$$

The compatible basis is $C = (1)$, the jump degree is $j_1 = 2$, and the full normal family is $(1, \tau)$. A full coordinate vector $(x_0, x_1)^T$ represents the class of $x_0 + x_1\tau$.

Since $N_\Phi = N_\Psi = 0$, the map $T$ vanishes. Thus

$$M = 2, \qquad r = 0, \qquad h = 1, \qquad M_0 = 1,$$

and the $\mathrm{Ext}_{0,\tau}$ coordinate family is $(\tau)$.

### The six outputs

$$\Pi_t = \begin{pmatrix} \theta & 0 \\ \tau & \theta + \tau^2 \end{pmatrix}, \quad \Pi_{0,t} = \begin{pmatrix} \theta + \tau^2 \end{pmatrix},$$

$$K = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 1 \end{pmatrix},$$

$$P = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 \end{pmatrix}.$$

In particular,

$$t \cdot \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = \begin{pmatrix} \theta x_0 \\ x_0^q + \theta x_1 + x_1^{q^2} \end{pmatrix}.$$

The map $P$ retains the constant coordinate, and $K$ inserts an $\mathrm{Ext}_{0,\tau}$ coordinate as $(0, y)^T$.

### Why the entry $\tau^2$ occurs

For the second coordinate, begin with $V = c\tau$. Its $t$-multiple is

$$\Psi_t(c\tau) = \theta c\tau + c^q\tau^2.$$

The inner change associated to the constant $H = c^q$ is

$$H\Phi_t - \Psi_t H = c^q\tau^2 - c^{q^2}\tau.$$

Subtracting it leaves $(\theta c + c^{q^2})\tau$. Hence the second diagonal entry of $\Pi_t$ is $\theta + \tau^2$.

### The exact sequence and the auxiliary matrix R

Here $Q_t = (\theta)$ and

$$0 \longrightarrow (\mathbb{G}_a, \theta + \tau^2) \xrightarrow{\binom{0}{1}} (\mathbb{G}_a^2, \Pi_t) \xrightarrow{(1 \;\; 0)} (\mathbb{G}_a, \theta) \longrightarrow 0.$$

Although $RK = 1$, in this example

$$R\Pi_t = (\tau \;\; \theta + \tau^2) \neq (0 \;\; \theta + \tau^2) = \Pi_{0,t}R.$$

Thus the auxiliary matrix $R$ is a coordinate left inverse of the inclusion, but is not itself a $t$-module retraction.

# 7. Matrix examples and the partial basis issue

## The pair with $J_2$ in both constant parts

Let $q = 3$ and

$$J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \Phi_t = (\theta + \tau^2)I_2 + J_2, \qquad \Psi_t = (\theta + \tau)I_2 + J_2.$$

Here $C = I_2$, $j_1 = j_2 = 2$ and $M = 8$. For $H = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$T(H) = HJ_2 - J_2H = \begin{pmatrix} -c & a - d \\ 0 & c \end{pmatrix}.$$

Therefore $r = 2$ and

$$\ker T = \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} : a, b \in L \right\}, \qquad h = 2, \qquad M_0 = 6.$$

The auxiliary matrix $S$ is

$$S = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

It has only one zero: $E_{12}$ is the only elementary matrix in $\ker T$. The additional kernel direction $I_2 = E_{11} + E_{22}$ is a linear combination of directions whose individual images under $T$ are nonzero. Counting zeros of $S$ would therefore give the wrong quotient dimension.

A basis of $\operatorname{im} T$ can be taken as $E_{12}$ and $-E_{11} + E_{22}$. The $\mathrm{Ext}_{0,\tau}$ basis combines these two constant directions with the four positive directions $E_{is}\tau$. In the quotient coordinates used by the checked output,

$$Q_t = \begin{pmatrix} \theta & 0 \\ 1 & \theta \end{pmatrix}.$$

The quotient has underlying additive group $\mathbb{G}_a^2$, with a nonscalar $t$-action. Different valid basis choices can change the displayed matrices while representing isomorphic coordinate descriptions.

## An upper-right entry $1 + \tau$

Another admissible input at $q = 3$ is

$$\Phi_t = \theta + \tau^2, \qquad \Psi_t = \begin{pmatrix} \theta + \tau & 1 + \tau \\ 0 & \theta + \tau \end{pmatrix}.$$

Set the matrix sizes to $d = 1$ and $e = 2$. Here $N_\Phi = 0$, $N_\Psi = J_2$, and

$$T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -b \\ 0 \end{pmatrix}.$$

Thus $M = 4$, $r = 1$, $h = 1$ and $M_0 = 3$. A normal representative is in $\mathrm{Ext}_{0,\tau}$ exactly when its constant column lies in the span of $(1, 0)^T$. The positive off-diagonal term $\tau$ affects $\Pi_t$ and $\Pi_{0,t}$; the constant obstruction $T$ uses only $N_\Phi$ and $N_\Psi$.